

# A Practical Scaling Laws Guidance Note for Actuarial Ratemaking

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*Companion guidance to  
Scaling Laws, Tabular Data and Actuarial Ratemaking Models*

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## Purpose

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This note summarizes the practical implications of the paper *Scaling Laws, Tabular Data and Actuarial Ratemaking Models*, available on SSRN at [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=6073948](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6073948). The paper is accompanied by a supplementary document containing a reproducible workflow on the public freMTPL2 dataset (Dutang & Charpentier, 2026; Wüthrich & Merz, 2023), and an open-source GitHub repository at <https://github.com/RonRichman/frmtpl-scaling-laws> containing the source code, data-preparation script, reference outputs, and a Colab notebook. Readers adapting the protocol described in this note to internal data may find the repository particularly useful as a starting point.

The note is intended for pricing actuaries, actuarial managers, model governance teams, and research stakeholders who want to understand what the findings imply for model development practice.

The paper investigates whether neural scaling laws, empirical regularities first established for large language models, also apply to actuarial ratemaking (Kaplan et al., 2020; Hoffmann et al., 2022). Using a large motor insurance dataset, it shows that held-out Poisson deviance improves predictably as data, model capacity, and compute increase, and that different model families convert these resources into better fit at different rates. The paper does not argue for abandoning GLMs or for putting large neural networks directly into rating plans.

Two implications follow for actuarial teams. First, scaling laws can be used to predict the practical limits of parameter scaling within a model family, that is, the point at which adding capacity stops translating into better held-out fit. Second, when these relationships are estimated locally, model capacity becomes a planning variable that responds to changes in available data: more rows justify testing larger models, but only by an amount the scaling relationships quantify. As labelled experience grows, the case for maintaining challenger models, comparing them with deployed models, and using their diagnostics for monitoring, segmentation, and model development strengthens.

## Executive Summary

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- GLMs remain essential because they are transparent, stable, familiar to regulators, and difficult to beat in small-data settings.
- GBMs, including LightGBM, XGBoost, and CatBoost (Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018), are strong tabular baselines that can serve as a primary benchmark in any scaling-law evaluation alongside the GLM. The paper’s empirical study did not cover GBMs because the existing model suite was already complex and adding a tree-based family with its own preprocessing requirements was a step too far for a single study, but the broader tabular machine-learning literature is consistent in finding that GBMs scale well with data on medium-sized tabular tasks (Grinsztajn et al., 2022; McElfresh et al., 2023).
- Scaling-law analysis helps pricing teams decide when richer models are worth testing, rather than treating model size as a guess.

- On the studied multi-million-row motor frequency task, TabM and TabM-mini (Gorishniy et al., 2024) make the strongest use of additional data and parameters among the models tested, indicating they may be most attractive in large-data regimes.
- Increasing the size of a supervised tabular Transformer is not, by itself, sufficient: architecture, inductive bias, and training objective also matter.
- The most realistic near-term use of high-capacity models in regulated environments is as challenger and monitoring models, even where filing constraints prevent direct deployment.
- Small deviance improvements can correspond to better likelihood assigned to emerging claim counts and to materially different interaction patterns, and therefore warrant inspection rather than dismissal.

## Scaling Laws In Actuarial Practice

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Pricing actuaries already reason in terms of credibility: more relevant experience supports more refined estimates. Scaling laws extend this reasoning empirically. They estimate how held-out loss changes as three resources increase: labelled training data, model capacity, and training compute.

A scaling law in this context is a simple empirical relationship of the form

$$L(N) \approx L_\infty + A_N N^{-\alpha},$$

where  $L$  is held-out loss (in this paper, mean Poisson deviance),  $N$  is the number of training rows,  $L_\infty$  is an empirical performance floor over the observed range,  $A_N$  is a fitted constant, and  $\alpha$  (alpha) is the data-scaling exponent. The exponent  $\alpha$  captures how quickly held-out loss falls toward the floor as more rows are added: a larger value means faster improvement. Parallel forms hold for parameter scaling and compute scaling,

$$L(P) \approx L_\infty + A_P P^{-\beta}, \quad L(C) \approx L_\infty + A_C C^{-\gamma},$$

where  $P$  is the number of trainable model parameters,  $C$  is a training compute proxy, and  $\beta$  (beta) and  $\gamma$  (gamma) are the corresponding exponents. Within the fitted range of an experiment, these exponents allow simple predictions about how much improvement to expect from more data, more parameters, or more compute.

This empirical form supports more systematic answers to practical questions:

- Has the current model family saturated?
- Is another year of data sufficient to justify a larger challenger model?
- Are extra parameters being converted into better held-out fit?
- Are model gains stable enough to justify governance work?
- Does the challenger model reveal credible interactions missing from the filed model?

The practical output is not the fitted exponents themselves but the planning structure they support. When experience grows, the team has a basis for testing specific larger models, comparing them with the current benchmark, and inspecting where the indications change. The exponents make these planning decisions quantitative rather than judgmental.

## A Practical Model-Choice Map

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The table below gives a practical reading of the empirical results, supplemented by what is known from the broader tabular-learning literature. The row cutoffs are indicative, not universal rules.

| Data regime                     | Recommended modelling posture                                                                                                                                               | Practical governance posture                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Small labelled data             | GLMs and regularized regressions as the main tools. Treat larger neural models cautiously. GBMs are usable but offer limited gains over a well-specified GLM at this scale. | Focus on stability, credibility, transparent effects, and simple validation.                                                   |
| Mid-size data                   | Maintain the GLM as a benchmark; test a tuned GBM as the primary machine-learning challenger and a compact neural challenger as a secondary option.                         | Use challengers to find missed interactions, stress-test segmentation, and monitor drift.                                      |
| Large labelled data             | Test GBMs and TabM/TabM-mini-style models. Compare both against the GLM benchmark.                                                                                          | Use champion-challenger reporting, seed stability, calibration checks, and segment diagnostics before any deployment decision. |
| Very large or fast-growing data | Treat model capacity and compute as budgeted research resources. Consider self-supervised representation learning on quote or policy data.                                  | Separate the best model for risk understanding from the model that is ultimately filed or deployed.                            |

## Where GBMs Fit

The paper’s empirical work focuses on GLMs, feedforward networks, tabular Transformers, and TabM/TabM-mini (Gorishniy et al., 2024). GBMs, including LightGBM, XGBoost, and CatBoost (Chen & Guestrin, 2016; Ke et al., 2017; Prokhorenkova et al., 2018), were not included. The reason is practical rather than dismissive: the paper already involved a complex suite of neural models sharing a common embedding-based feature representation, and adding a tree-based family that benefits from raw numerical inputs and its own preprocessing pipeline was a step too far for a single study. A clean direct comparison between GBMs and the neural models on a common feature representation was therefore not feasible within the scope of the paper.

This omission does not change the importance of GBMs in actuarial machine-learning practice. In broader tabular machine-learning evaluations, GBMs remain among the strongest baselines. Grinsztajn et al. (2022) report that tree-based methods often outperform standard neural networks on medium-sized tabular problems, with the gap attributed to inductive biases around uninformative features and irregular decision boundaries. McElfresh et al. (2023) qualify this with a larger benchmark in which differences are often modest with proper tuning. For most actuarial portfolios in the mid-data regime, a well-tuned GBM is among the strongest practical challengers to a GLM, and a scaling-law evaluation that omits GBMs as a benchmark will tend to overstate the gain available from neural models.

A note on applying scaling-law analysis to GBMs is in order. GBM capacity is governed jointly by the number of trees, tree depth, leaf count, and learning rate, and none of these forms a clean stand-alone parameter count analogous to  $P$  in the neural case. Defining a parameter-scaling exponent in the form used for neural models is therefore not straightforward. In practice, the size of a well-tuned GBM emerges from the hyperparameter optimization process: a deeper or wider configuration tends to fall out of standard tuning when more data justify it, rather than being selected through a deliberate scaling sweep. This note does not propose formal parameter-side scaling-law analysis on GBMs. Data-side analysis can still be useful as a descriptive summary of how a GBM benefits from additional rows, but the parameter question is best handled by the team’s standard tuning workflow rather than by analogy to the neural co-scaling rule.

Whether TabM-style models materially outperform a well-tuned GBM at multi-million-row scale on actuarial data is an open empirical question. The paper does not answer it. A pricing team running its own evaluation can include a tuned GBM as a primary benchmark alongside the GLM, and report the GBM-versus-GLM lift on the same scale as any neural-network result.

## Operational Toolkit

This section gathers the tools needed to act on the findings. The first subsection gives the planning math and worked examples; the second turns that math into a controlled study; the third sets out a recurring annual workflow.

### Sizing The Next Challenger

The scaling-law machinery answers two practical planning questions: *how much improvement should we expect from another year of data?*, and *what model size should we test next?* Both reduce to short calculations once the relevant exponents are estimated locally. The exponents  $\alpha$  (data) and  $\beta$  (parameters) are introduced in the previous section.

**Background constants.** For TabM/TabM-mini, the paper reports a data exponent of approximately  $\alpha \approx 0.41$  and a parameter exponent of approximately  $\beta \approx 0.15$  in the headline summary. The ratio is therefore

$$\frac{\beta}{\alpha} \approx \frac{0.15}{0.41} \approx 0.36.$$

These constants come from the best-achieved data envelope in the main paper. A complementary bivariate Chinchilla-style fit gives a similar ratio of about 0.40, and the more conservative main-sweep TabM exponent gives  $\beta/\alpha \approx 0.48$ . The qualitative implication is the same across all three: along the fitted local frontier, parameters scale faster than data, but labelled-data growth still matters.

**Worked example 1: forecasting next year's improvement.** The data-scaling form is  $L(N) - L_\infty \approx A_N N^{-\alpha}$ . Suppose a portfolio grows from 2.0M to 2.4M policy-period rows (a 20% increase) and the current reducible loss above the floor is approximately 0.0015. With  $\alpha \approx 0.4$ , the predicted reducible loss at the new data size is

$$0.0015 \times \left(\frac{2.0}{2.4}\right)^{0.4} \approx 0.0015 \times 0.93 \approx 0.00139.$$

The expected reduction is about 0.00011, or roughly 7% of the remaining gap to the floor. The number is small in absolute terms but sets the realistic expectation against which a richer challenger model is evaluated. If the challenger does not visibly outperform that data-only baseline on validation, the gain is not coming from the new architecture.

**Worked example 2: sizing the next challenger.** The same constants imply a co-scaling rule

$$N_\star \propto P_\star^{0.36},$$

where  $N_\star$  is labelled training rows and  $P_\star$  is model parameters along the fitted compute-efficient path. Parameters grow faster than labelled data, but labelled data still needs to grow. If parameters double, the heuristic implies a labelled-row increase of

$$2^{0.36} \approx 1.28,$$

or about 28%. Inverting the rule for annual planning, a 20% increase in labelled data implies

$$P_\star \propto N_\star^{1/0.36}, \quad 1.20^{1/0.36} \approx 1.66,$$

that is, a TabM/TabM-mini challenger with roughly 60 to 70% more parameters than the current size. A reasonable refresh tests one or two larger variants around  $1.5\times$  to  $2\times$  the current parameter count alongside the existing champion and challenger.

**Worked example 3: reading deviance on the likelihood scale.** Suppose a challenger improves mean Poisson deviance by  $\Delta D = 0.0020$  on a validation portfolio of 20,000 policy-period rows. Since Poisson deviance differs from twice the negative log-likelihood by a constant that does not depend on the model, the total log-likelihood improvement assigned to the observed claim-count vector is

$$20,000 \times \frac{0.0020}{2} = 20.0$$

on the natural logarithmic scale. This is a model-fit statement: the challenger assigns a materially higher likelihood to the observed counts. It is still not a profit estimate, because pricing value also depends on severity, calibration, expenses, margins, retention, competitive response, and governance constraints.

**Worked example 4: setting an annual validation hurdle.** In worked example 1, the extra 20% of labelled rows was expected to reduce mean deviance by about 0.00011 through data growth alone. If a larger challenger improves validation deviance by only 0.00004 relative to the retrained old-size challenger, and seed-to-seed variation is of similar size, the result is weak evidence for increasing model size. If the larger challenger improves deviance by 0.00020 across repeated seeds and shows stable calibration, the gain is larger than the data-only expectation and is more credible as an architecture or capacity benefit.

**Caveat on the compute assumption.** The Chinchilla-style allocation assumes that compute is also scaled jointly with data and parameters. A team that receives more data without adjusting its compute budget cannot apply the rule mechanically. A more defensible approach is to re-estimate  $\alpha$  and  $\beta$  locally and choose the model size that beats the production benchmark on validation, irrespective of what the formal optimum prescribes. The rule is a search heuristic, not a production directive.

The rule above is specific to TabM-style neural models. As discussed in the previous section, a clean parameter-scaling exponent for GBMs is not straightforward to define, and the parameter-sizing rule does not transfer directly to that family.

## Running A Scaling-Law Evaluation

The paper provides a calibration point, not a set of universal constants. A commercial actuarial team can treat the empirical findings as hypotheses to test locally. The relevant question is: on this portfolio, with this target, this feature set, this validation design, and this governance environment, how much benefit does additional data, model capacity, and compute deliver?

Reasonable starting expectations, based on the paper, are the following:

- GLMs remain strong baselines, particularly when data are limited.
- Additional labelled rows help most model families, although the rate of improvement differs by family.
- Larger models help only when the architecture and training objective convert extra capacity into useful signal.
- GBMs typically scale well with data on medium-sized tabular problems and remain competitive in many regimes, so they can be benchmarked rather than assumed to be dominated by neural models.
- TabM-style models become more attractive in the multi-million-row regime, but their dominance over a tuned GBM on actuarial data is not established.
- Small deviance gains warrant calibration and segment-level diagnostics before being dismissed.

A practical evaluation can be run as a controlled model-capacity study:

1. **Choose a clean target.** Start with frequency, using an exposure-scaled Poisson objective. Defer severity and pure premium unless the team has sufficient claim-level data to study them credibly.
2. **Create a fixed modelling extract.** Use stable feature definitions, documented exclusions, and fixed learning, validation, and test samples. Include an out-of-time validation period where possible.

3. **Create nested training fractions.** For example, train on 10%, 25%, 50%, 75%, and 100% of the learning data, with smaller samples nested inside larger samples. This isolates the effect of adding experience.
4. **Choose model families.** Include the current production model or filed-model proxy, a GLM benchmark, a tuned GBM (LightGBM, XGBoost, or CatBoost) as the primary tabular machine-learning challenger, one standard neural baseline, and one strong tabular neural model such as TabM/TabM-mini (Gorishniy et al., 2024).
5. **Choose a modest capacity ladder for the neural families.** Use small, medium, and large variants. The first study can aim to estimate trends and saturation points rather than to search for a leaderboard optimum. For GBMs, capacity is driven by hyperparameter tuning, so a separate scaling ladder is not required; report the tuned GBM as a single benchmark configuration.
6. **Repeat random seeds.** Use at least three seeds for neural models, and preferably five for an internal study. Report both the average result and seed variability. For GBMs, repeat with different bagging seeds where the implementation supports it.
7. **Fit descriptive scaling summaries.** For each neural family, estimate data scaling from loss versus row count and parameter scaling from loss versus model size. For the GBM, fit data scaling on the same form to summarize the benefit of additional rows; do not attempt a parameter-side fit.
8. **Evaluate actuarial relevance.** Compare deviance, log-likelihood lift, calibration, lift, segment-level indications, interaction diagnostics, and stability across seeds and validation periods.
9. **Turn the result into an action.** Decide whether the richer model can remain a monitor, inform production-model refinements, justify a larger annual challenger, or enter a formal governance pathway.

The result is a controlled experiment that replaces informal capacity searches with measured estimates of the return on additional data, capacity, and compute.

A reproducible compact version of this protocol on the public freMTPL2 dataset (Dutang & Charpentier, 2026; Wüthrich & Merz, 2023) is available in the accompanying GitHub repository at <https://github.com/RonRichman/frmtp2-scaling-laws>. The repository includes a Colab notebook, source code, data-preparation script, reference outputs, and figures, and is intended to let practitioners inspect and rerun the workflow before adapting it to internal data.

| Commercial question                          | Scaling-law evaluation output                                                                     |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|
| How much value did another year of data add? | Data-scaling curve and year-over-year validation comparison.                                      |
| Is the current model family saturating?      | Flattening data or parameter scaling slope, plus unchanged validation lift.                       |
| Should we test a larger challenger?          | Parameter-scaling slope, seed stability, and the local data-growth heuristic.                     |
| Which model family uses our data best?       | Family-specific scaling curves and held-out calibration diagnostics.                              |
| Is the GBM saturating before the GLM?        | Comparison of GBM and GLM data-scaling curves; GBMs can flatten earlier than expected.            |
| Are we missing interactions?                 | Challenger-versus-benchmark segment tables and interaction heatmaps.                              |
| What should management fund next?            | A recommendation on data acquisition, compute budget, model-family testing, or governance review. |

### The Annual Refresh Workflow

When another year of experience emerges, a structured workflow could include the following steps:

1. Refresh the modelling extract and confirm that exposure, claim emergence, and feature definitions are comparable with the prior extract.
2. Retrain the current production benchmark or filed-model proxy.

3. Retrain the GBM challenger at its current configuration.
4. Retrain the current neural challenger at its current size.
5. Use the sizing heuristic to choose one or two larger neural challenger sizes to test.
6. Evaluate all models on a fixed validation or out-of-time holdout.
7. Compare deviance, calibration, lift, segment-level indications, and seed stability.
8. Decide whether the richer model is only a monitoring improvement, a source of GLM refinements, or a candidate for deeper governance review.

This workflow turns each data refresh into a structured re-evaluation rather than a re-fit of the existing model. It also produces the year-over-year comparisons that make scaling-law claims falsifiable on the portfolio's own terms: if the predicted improvement does not materialize, the local exponents can be re-estimated rather than the rule being abandoned.

## Governance and Deployment Considerations

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This section addresses where richer models fit in governance, monitoring, and deployment, and how compute enters planning.

### A Governance Ladder For Richer Models

The findings have practical use even where direct deployment is not feasible. In many North American rate-filing contexts, a large neural model is difficult to file directly. This does not eliminate its diagnostic value.

A practical governance ladder has the following levels:

1. **Production benchmark:** maintain the GLM or current filed model as the transparent production anchor.
2. **Internal challenger:** train a stronger model (typically a GBM in the mid-data regime, or a TabM/TabM-mini in larger-data settings) on the same modelling extract and compare it with the production benchmark.
3. **Monitoring model:** use the challenger to detect calibration drift, emerging adverse selection, or missed interactions.
4. **Model-improvement input:** translate stable challenger findings into simpler filed-model refinements, such as interactions, territorial reviews, segmentation changes, or underwriting monitoring.
5. **Deployment candidate:** consider direct use only where the governance, explainability, filing, systems, and monitoring requirements can be satisfied.

The model used to understand the risk does not have to be the same as the model filed as the rating algorithm. A complex challenger can produce actuarial evidence even when the filed solution must remain simpler.

### Maintaining A Best-Available Internal Model

It can be useful for a pricing team to maintain the strongest predictive model it can responsibly build, even if that model is not deployed. The purpose is not to replace professional judgment but to provide a stronger reference against which actual experience can be monitored.

The best model on hand can address questions such as the following:

- Whether the production model is leaving systematic signal unexplained.
- Which segments show the largest difference between filed-model and challenger indications.
- Whether differences are stable across time, random seeds, and validation samples.
- Whether the differences suggest a missing interaction or a change in portfolio mix.

- Whether the challenger detects deterioration before it appears in standard monitoring reports.

### Compute As A Planning Input

Pricing-model planning has traditionally focused on data, variables, credibility, and governance. Scaling-law results add compute and model capacity to that planning conversation.

This does not imply spending compute indiscriminately. It implies that a team can usefully know what compute is required to test the next relevant model size, how long that run takes, whether the infrastructure is reproducible, and whether the expected gain is material relative to the cost.

Larger insurers may have a scale advantage here: more policy-period rows, more quote data, more claims history, and better infrastructure can support richer monitoring and challenger-model programs. Smaller insurers may rely more on simpler robust architectures, pooled public benchmarks, or vendor and industry tools while retaining standard governance and validation requirements.

### Interaction Discovery And Business Interpretation

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Small average deviance gains can be easy to dismiss. The more useful question is where the model changes the indicated frequencies.

The supplementary FMTPL reproduction provides a concrete example. The lower-deviance TabM-mini model produces a visibly different interaction surface from the GLM between driver age and bonus-malus status. The differences are specific and inspectable: TabM-mini indicates materially *higher* expected frequencies than the GLM for drivers aged 70+ with bonus-malus 51-75 and for drivers aged 18-24 with bonus-malus above 100, and materially *lower* frequencies for drivers aged 18-24 with bonus-malus 51-75 and for drivers aged 40-49 with bonus-malus 76-100. Several of these cells are too thin in the public FMTPL data to act on directly, but they are the kind of interaction signal a pricing team would investigate further on a credible production portfolio.

This is the practical content of the deviance improvement: not only a lower aggregate number, but disagreement with the GLM in identifiable cells that an actuary can investigate. In practice, the challenger can be used to produce diagnostics such as the following:

- one-way relativities compared with the GLM;
- two-way interaction heatmaps;
- calibration by exposure-weighted cells;
- top segments where challenger and production indications differ;
- stability of those differences across seeds and validation periods.

The actuarial question is not only whether the score is lower. It is whether the model is finding stable, credible structure that improves understanding of the risk.

### When Bigger Models Are Not Worth It

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Bigger models do not need to be tested or promoted mechanically. They are usually not worth the effort when:

- labelled data are limited or unstable;
- validation gains are small and not stable across seeds;
- the gain appears only in cells with weak exposure;
- the model is poorly calibrated even if deviance improves;
- the team cannot explain the material changes in indications;

- the model cannot be monitored or reproduced reliably;
- there is no operational path to use the result.

In these cases the appropriate response is usually to improve data quality, strengthen the validation design, refine the GLM, introduce targeted interactions, or test a different architecture rather than simply increasing size.

## Data Quality And The Limits Of Additional Rows

Adding rows is not the same as adding useful information. A new year of experience is valuable only if exposure is credible, claim emergence is sufficiently mature, feature definitions are stable, and portfolio mix changes are understood.

For frequency modelling, every policy-period contributes a claim count, which makes scaling-law estimation relatively well-behaved. Severity and pure-premium modelling will likely behave differently, and the empirical exponents reported in the paper should not be assumed to carry over. Severity is observed only on the smaller subset of policies that produced a claim, the loss surface is heavy-tailed and noisier, and the same volume of policy-period rows translates into far fewer effective training examples. The protocol described above transfers directly to severity modelling (nested training fractions, seed averaging, family-wise capacity ladders, calibration diagnostics), but the fitted  $\alpha$  and  $\beta$  should be re-estimated locally, and saturation may appear at a much smaller effective sample size than for frequency. Pure-premium modelling adds the further complication of combining frequency and severity exponents under a single loss function.

Quote and policy data without fully emerged outcomes may also become relevant. The analogy with foundation models is not that insurance records are text, but that both settings contain large volumes of structured information before the final supervised label is available. In language modelling, a model can first learn useful representations from next-token or related self-supervised tasks on broad corpora, and can then be adapted to downstream tasks once task-specific labels or instructions are available (Devlin et al., 2018; Kaplan et al., 2020; Hoffmann et al., 2022). In insurance, the analogous idea would be to learn representations from quote, policy, endorsement, and exposure records before claims have fully emerged, and then connect those representations to frequency, severity, retention, or monitoring tasks once outcome labels are credible.

## What To Report To Management

A compact management report after each major model refresh could include the following items:

| Report item                             | What it answers                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------|
| Champion and challenger deviance        | Did the richer model improve held-out statistical fit?                                       |
| Log-likelihood lift                     | How large is the improvement on the natural likelihood scale?                                |
| Calibration diagnostics                 | Is the challenger well calibrated overall and by important segments?                         |
| Seed stability                          | Is the result robust to training randomness?                                                 |
| Segment differences                     | Where do indications change most materially?                                                 |
| Interaction diagnostics                 | Which two-way or higher-order patterns appear to be missed by simpler models?                |
| GBM versus neural-challenger comparison | Is the additional complexity of a neural challenger justified relative to the GBM benchmark? |
| Governance recommendation               | Should the model remain a monitor, inform GLM refinement, or enter formal deployment review? |

The objective is not to celebrate a lower loss number in isolation. It is to determine whether the lower loss corresponds to stable, credible, governable changes in risk indications.

## What Not To Overclaim

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The findings should be read with appropriate limits.

- The main evidence comes from one large motor frequency dataset.
- The conclusions may differ for other lines, jurisdictions, feature sets, validation designs, and targets.
- The paper does not prove that all ratemaking datasets obey the same exponents.
- The study is not a full benchmark of every tabular model family. In particular, GBMs were not in the main experiment, and a comparison of TabM-style models against a tuned GBM at production scale on actuarial data remains open.
- Poisson deviance gains are statistical model-fit gains, not direct profit estimates.
- Rate filing, model governance, and business adoption require additional evidence.

These limits do not weaken the practical value. They clarify how the results can be used: as a disciplined guide to experimentation, monitoring, and capacity planning.

## Bottom Line

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The practical implication of the paper is that actuarial model development can become more systematic about scale. When data grows, the appropriate response is not simply to refit the existing model. The relevant questions are whether the data growth justifies a larger challenger, whether the current architecture is saturating, and whether a richer model reveals credible structure missed by the production benchmark.

GLMs remain the central deployed model in most jurisdictions. GBMs are strong tabular challengers and can be benchmarked routinely. Larger neural models warrant deployment only after meeting validation, calibration, stability, and interpretation standards. In large-data regimes, a best-available challenger model is more readily justified as a standard actuarial control than as a research luxury.

More broadly, these results imply treating model size as a third planning variable alongside data and compute, rather than as a property determined by family choice (a “GLM shop” or a “GBM shop”) or settled by architectural preference. The operational tools described above (the sizing heuristic, the controlled evaluation, the annual refresh) follow from this view.

## References

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- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining (kdd)* (pp. 785–794). doi: 10.1145/2939672.2939785
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Dutang, C., & Charpentier, A. (2026). *French motor third-party liability datasets: fremtpl*. CASdatasets documentation. Retrieved from <https://dutangc.github.io/CASdatasets/reference/freMTPL.html> (Accessed 2026-04-30)
- Gorishniy, Y., Kotelnikov, A., & Babenko, A. (2024). *TabM: Advancing tabular deep learning with parameter-efficient ensembling*. Retrieved from <https://arxiv.org/abs/2410.24210>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems (NeurIPS)*, 35, 507–520.
- Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., ... others (2022). Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*.

- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., . . . Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., . . . Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems (neurips)*.
- McElfresh, D. C., Khandagale, S., Valverde, J., Ramakrishnan, G., Goldblum, M., Fowl, L., . . . Papyan, V. (2023). *When do neural nets outperform boosted trees on tabular data?* Retrieved from <https://arxiv.org/abs/2305.02997>
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). *CatBoost: Unbiased boosting with categorical features*. Retrieved from <https://arxiv.org/abs/1706.09516>
- Wüthrich, M. V., & Merz, M. (2023). *Statistical foundations of actuarial learning and its applications*. Springer Cham. Retrieved from <https://link.springer.com/book/10.1007/978-3-031-12409-9> (Open access; see Appendix B for the **freMTPL2** correction and cleaning) doi: 10.1007/978-3-031-12409-9